# FAN BELT DECISION:
# NEW VS. TOKUNBO
## (24 OCTOBER 2026)

**By: Marshal Bernard (MBA253193)**

### A. Decision and evidence

On Saturday 24 October 2026, while travelling from my school to Ibadan before my last class, my car (a Toyota Highlander) broke down on the road. A roadside mechanic discovered it was the fanbelt that broke and offered two options:

- Option 1: a new belt at ₦7,000, with an estimated life of six months
- Option 2: a tokunbo belt at ₦17,000, which could last up to 18 months if no other part interfered.

The breakdown for each option is as follows:

| | **Option 1 (New Belt)** | **Option 2 (Tokunbo Belt)** |
|---|---|---|
| **Belt** | ₦7,000 | ₦17,000 |
| **Transport** | ₦2,000 | ₦2,000 |
| **Workmanship** | ₦10,000 | ₦10,000 |
| **Total** | **₦19,000** | **₦29,000** |
| | | |

The figures derived were based on the mechanic's verbal explanation. I could spare up to ₦20,000 for this repair, without stretching. I chose the new belt in option 1 because I plan to sell the car within 30 days.

The payment was made in two installments with receipts attached:

- First installment: ₦9,000 (for belt and transportation, paid before work commencement)
- Second Installment: ₦10,000 (paid after work has been done and confirmed working).

### B. Budget constraint and utility-maximising choice

- x = number of new belts
- y = number of tokunbo belts
- $p_x$ = ₦7,000
- p_y = ₦17,000

Transport and workmanship (₦12,000) are the same under either option, so they are treated as a fixed cost (the mechanic quoted no labour difference). The budget constraint is:

$$7{,}000x + 17{,}000y \leq M - 12{,}000, \text{ where } M = ₦20{,}000.$$

The car needs exactly one belt, so the feasible bundles are (1,0) and (0,1). Both belts do the same job, so I represent preferences as:

$$U = ax + by,$$

where a and b are the months of service I expect to use from each belt. Since I will sell within about one month, and both belts outlast that, then:

$$a = b \approx 1, \text{ giving MRS} = a/b = 1.$$

The price ratio is:

$$p_x/p_y = 7{,}000/17{,}000 \approx 0.41.$$

Because MRS (1) exceeds the price ratio (0.41), each naira buys more utility on the new belt, so utility is maximised at (1,0). This matches my actual choice.

The result depends on my time horizon. If I were keeping the car, a = 6 and b = 18 (the best case) would give MRS = 0.33, below 0.41, favouring the tokunbo. At a more realistic 12 months, MRS = 0.50 and the new belt wins again.

**C. Counterfactual**

Suppose the new belt cost ₦18,000 instead of ₦7,000 (for example, only an imported brand in stock). The revised constraint is:

$$18{,}000x + 17{,}000y \leq M - 12{,}000.$$

The price ratio becomes 18/17 ≈ 1.06, which exceeds my MRS of 1. The tokunbo now gives more utility per naira, so I predict a switch to (0,1), for a total of about ₦29,000 (₦17,000 + ₦12,000). This is mainly a substitution effect: the new belt became relatively dearer, so I move to its substitute. The income effect is negligible, since a one-off belt is a small share of my overall spending and the good I would switch to did not change in price. This is my estimate only. What actually happened is that I bought the new belt at ₦7,000.

**D. Assumptions and managerial implication**

1. Perfect information and quality certainty. The model assumes I know each belt's true lifespan. In reality, the 6-month and 18-month figures came from the mechanic, who likely earns from the sale, and I could not verify them. If I doubted the quality of the new belt, I might have chosen the tokunbo despite my short horizon, to avoid a second breakdown while showing the car to buyers.

*2.* Time pressure and access costs. The model assumes I can freely compare all options at known prices. I was broken down on the road to Ibadan, so I could not shop around or check other prices, and I decided between the first two quotes. With time and lower search costs, I might have found a cheaper tokunbo, lowering p_y and possibly changing the choice.

- Managerial implication. Nigerian auto-parts dealers and mechanics could offer short written warranties or stated lifespans, such as a 3-month warranty on tokunbo belts and a 1-month warranty on new. This turns unverifiable claims into verifiable ones, builds trust, and lets customers with short horizons, such as car sellers, pay only for the service life they will actually use.

**E. Reflection and traceability**

The analysis depends most on my expected usage horizon (30 days) and the mechanic's lifespan claims, which determine a and b. The prices and total spent are backed by transfer receipts, but the tokunbo price and lifespans are verbal. Remaining uncertainty includes whether the sale happens within 30 days, whether the new belt really lasts six months, and whether buyers value a recently replaced belt. To test my explanation, I would (1) record the actual sale date, (2) get a second mechanic's quote for both belts, and (3) track the new belt's performance after six months.

OPay

Transaction Receipt

# ₦9,000.00

Successful

Oct 24th, 2026 13:27:58

| | |
|---|---|
| Recipient Details | **SEUN TUNDE LADEJOBI**<br>OPay \| 810****843 |
| Sender Details | **MARSHAL AKEM BERNARD**<br>OPay \| 806****424 |
| Transaction No. | 261024010100707214539705 |

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Transaction Receipt

# ₦10,000.00

Successful

Oct 24th, 2026 14:40:31

| | |
|---|---|
| Recipient Details | **SEUN TUNDE LADEJOBI**<br>OPay \| 810****843 |
| Sender Details | **MARSHAL AKEM BERNARD**<br>OPay \| 806****424 |
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